DF13426/Aprile Refree comments and response

XENON collaboration

July 2023

1 Major Changes to the Manuscript

We express our gratitude to both the referees for their time and comments. All their suggestions have been constructive in improving the quality of the manuscript. A diff file showing all the changes is included. The major changes have been described below:

- To address the second comment of Referee A, We have added a paragraph in Section 1 (Introduction), which present three physics cases to motivate particle searches at the keV level.
- As suggested by Referee B, We have added ANTARES in Tab 1. Further, We have updated the NOvA energy range based on their publication on the search for targeted supernova-like neutrinos. We have also updated the reference for Borexino from the arXiv article to the published version.
- **The biggest change is how various analyses are addressed now.** As suggested by Referee A, it was done to avoid any sort of confusion among readers. Further, section 2, which explains these analyses, has been drastically changed to systematically introduce the NR and ER channel and the three different modes within the NR channel. We have also added a paragraph in section 2 discussing acceptance for the analyses, which would be beneficial to understand Fig 1 (right).
- As suggested by both referees, We have modified the last paragraph of section 2 to explain why GW170817 is interesting.
- As suggested by Referee A, the discussion of the effective cross-section for ER is modified to explain the trends in Fig 1 (left) properly. Similarly, a few more sentences are added in the explanation for Fig 1 (right).
- As suggested by Referee B, the first paragraph of section 4 is modified to provide more details about deadtime to readers.
- As suggested by Referee A, the discussion of the p-value has been modified to make it clearer. Further, instead of "upper limit on the expected

number of background events", we call it "the minimum number of events corresponding to an excess" because upper limits have a specific statistical meaning in our field.

- **The other major change is that our 2nd limit is on ALPs instead of BSM particles.** This was one done because Referee A pointed out that the assumption that atomic electrons could fully absorb particle energies, would not be valid for particles other than ALPs. We have also added a sentence on axio-electric effect.

We have also replied to the concerns of referees in-line in their reports. Our replies are in blue text.

2 Report of Referee A

The XENON collaboration carried out an interesting search for particle signatures in coincidence with a few gravitational wave detection events. Taking advantage of the detector's low background rate and low energy threshold, this study substantially extends the search to lower energy regions than what has been done by other experiments. This topic is suitable for publication on PRD, but I would like to see my following concerns addressed first.

First, the paper appears to have been written with the direct detection dark matter community as the targeted audience, but I would expect this study to be interesting to people in cosmology as well. For readers outside the direct detection field, it is not straightforward to understand the difference between the NR, ER, CEvNS and S2-only channels, so some more descriptive names are needed. Actually from my understanding, these are not different signal channels of the detector as referred to, but different analysis modes.

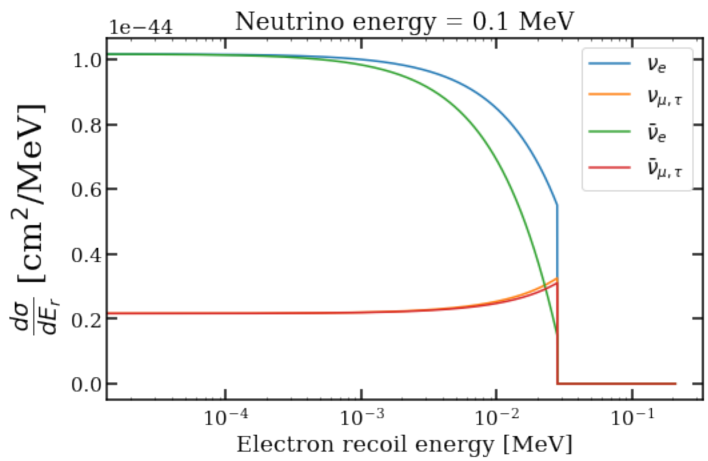
We have changed section 2 to systematically introduce the NR and ER channel and the three different modes within the NR channel. We have also changed these names to make it more intuitive. We have edited the whole manuscript to reflect these changes.

Second, the typical energy level associated with astrophysical events is MeV (nuclear energy level). It will be helpful if the authors can present some physics cases that motivate particle searches at the keV level out of the mergers.

We have added a paragraph in introduction (section 1) to address this. We have mentioned 3 physical cases which can possibly lead to detection of particles in keV level.

Third, the paper makes an assumption that sterile neutrino energies could be fully absorbed by atomic electrons, which appears odd. Can the authors elaborate on the BSM physics that could allow such interactions?

Yes, the assumption and hence the limit are valid only for ALPs. We have edited the paper accordingly. We have also added a single-sentence description of axio-electric effect before equation 4 and attached a reference that goes into detail.



Fourth, the local vs global p-value discussions caught me a bit. I think the authors need to reexamine the definition and discussion. If a non-conventional definition is used, it would need some additional explanation.

We have rewritten the discussion to make it clearer. Accounting for the look-elsewhere effect is common in various fields, We have added a reference that goes into the details. Similarly, Sidak correction is a standard way to correct for the look-elsewhere, as also mentioned in the previous reference.

Additional comments:

- Fig 1. Left: it is not obvious to me why the effective neutrino-electron elastic scatter cross section would have a sharp turn at 360keV (the text says 36keV which should be a typo) – Wewas expecting the energy window cutoff to bring in a gradual transition. I also don't understand why it would be almost flat between 360keV and 10MeV – does the 210keV energy cutoff magically cancel the increase of physical v-e cross section? This is not impossible but would be a rare coincidence.

  We have rewritten the explanation of Fig 1. Left in the text to make it clearer, and corrected the typo. There is no considerable change of physical v-e cross section with neutrino-energy at any energies, making the effective cross-section flat between 360 keV and 10 MeV. The effective cross section decreases because lower energy neutrinos are incapable of generating ERs throughout the entire energy range of the ER channel. As can be seen in the attached figure, to satisfy the kinematics of the system, the cross section has to go sharply to 0 at a certain recoil energy for a given neutrino energy. That's the reason for a sharp turn in the effective cross section.

- Fig 1 right: why does the S2 only analysis mode have a higher effective cross section than CEvNS for neutrino energies above 25MeV? According to Table II, the S2 only search has 10 times smaller target mass; the energy threshold is marginally better but this should only help the low-energy neutrino case and not for high energy neutrino interactions.

The reason is the lower signal acceptance of 2-fold. We have added a paragraph in section 2 discussing acceptance for the analyses. We have also expanded the second-last paragraph of section 3, to explain the behaviour of the Fig 1 (right).

- The authors explain that they focus on GW170817 because of its distance to Earth, but Wethink it is more important that it is a NS-NS merger and the only GW event with co-detection of electromagnetic signatures.

  We have modified the last paragraph of section 2 to add these points.

3 Report of Referee B

This paper presents limits on the fluxes of neutrinos down to 17 keV that could possibly be associated with gravitational wave events. Even if not theoretically motivated, the novel extension of dark matter search in this way to astrophysical neutrino searches is a useful result. The paper is generally well-written. But the authors need to provide some missing information for the results to be publishable.

Section IV: after "Accounting for deadtime due to various cuts and background events," it is not clear what was done. In fact the information about the livetime per GW is not sufficient for understanding how to interpret the results.

The standard +/-500 second time window was originated in Baret et al. Astropart.Phys. 35 (2011), 1, to safely cover a wide range of emission scenarios. But all times within the window are not equal. For example if a search is missing 30 seconds out of the 1000 second total, it does not matter a lot if those 30 seconds occur at the end of the time window, but it is a severe degradation if those 30 seconds are centered on the GW.

So the information provided, that the average livetimes are 935 seconds for the NR channel, going down to 650 seconds for the S2-only channel, needs to be supplemented in order for the limits to be meaningful. Where are the gaps in each case? Especially S2-only channel. And it is not clear how much variation from GW to GW occurs, since only an average is provided. It is not clear how well the central time around each GW is actually covered.

The authors need to provide more details, for the reader to judge if the limits are actually constraining.

We have added the mentioned reference. Further, We have added more details about the deadtime. The deadtime is due to cuts or background signals, and is uniformly distributed in time throughout the window. We have also added the range of livetime for different GWs.

In Section IWe"Out of these GWs, the particle flux is dominated by GW170817 because the flux is inversely proportional to the square of distance." and in Section V "We are most sensitive to the merger of neutron stars – GW170817, due to its close proximity." It is reasonable to highlight the results for GW170817.

But what is said here to motivate it is not completely correct. In the first sentence, there must be an assumed model (either the same for both and assuming isotropic emission, or two separate models) in order to make a statement as to which GW gives a higher detectable flux at Earth. Since there is no assumed model in the analysis, this statement is neither true nor false. Similar for the second sentence. However, it is fine to say the GW170817 result is the most interesting and is highlighted, because of its proximity, and because it was a merger of neutron stars and detected in light, not just GW.

We have modified the last paragraph of section II according to your suggestion. The sentence in section V has also been modified.

One should also consider to mention that all the constrained neutrino fluxes correspond to very high fluxes, generally exceeding the mass of the combined system if isotropic emission is assumed.

We have added the information in section 5 (at the end of 2nd paragraph).

"The subsequent observation of electromagnetic counterparts from binary neutron star coalescence in 2017 [3]" Here should include a citation to the GW170817 multimessenger paper, Astrophys.J.Lett. 848 (2017) 2, L12

We have cited this reference.

For completeness in Table 1 the ANTARES searches should be included, e.g. A. Albert et al JCAP 04 (2023) 004

We have included it.

Note there is a typo in Pierre Auger, "Observatiory"

We have fixed it.

Ref. 10 is now published, Eur. Phys. J. C (2023) 83:538

We have updated the reference.